

PROJECT REPORT

Hotel Reservation and Billing System

The Grand Aurelia Hotel

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1. Abstract

This project is a console-based hotel reservation system written in Python. It shows available rooms, lets a guest pick a room by number or type, collects guest details, adds optional services, calculates a bill with 18% GST, records the payment method, and confirms or cancels the booking. It uses dictionaries, lists, functions, loops, conditionals, and exception handling.

2. Introduction

Manual hotel booking is slow and error-prone. Automating room selection and billing reduces mistakes and saves time. This project is a small model of such a system, built with only core Python.

3. Problem Statement

Build a program that manages room availability, records guest and booking details, and produces an accurate final bill with tax.

4. Objectives

- Show only available rooms
- Prevent double booking
- Validate guest count and inputs
- Add optional services
- Generate a correct itemised bill with GST
- Support payment selection and booking confirmation or cancellation

5. Tools and Technologies

Item	Detail
Language	Python 3.6 or above
Libraries	None (built-in only)
Interface	Command line
Editor / IDE	Any (VS Code, IDLE, etc.)

6. System Design

6.1 Data Structures

- rooms (dictionary): key is the room number (string); value is a list [type, price, max_guests, balcony, bed, view, status].
- booking (dictionary): stores room, name, phone, guests, nights, and request.
- services (list): each entry is [service_name, price].

6.2 Room Data

Room	Type	Price/Night	Max Guests	Bed	View
101	Single	Rs. 1200	1	Single Bed	Garden View
102	Single	Rs. 1200	1	Single Bed	Garden View
103	Double	Rs. 1800	2	Queen Bed	City View
104	Double	Rs. 1800	2	Queen Bed	City View
105	Deluxe	Rs. 2500	2	King Bed	Garden View
106	Suite	Rs. 3500	3	King Bed	Premium View

6.3 Services

Service	Price
Breakfast	Rs. 250
Food	Rs. 300
Laundry	Rs. 200
Room Service	Rs. 150

6.4 Modules (Functions)

Function	Purpose
show_rooms()	Prints a formatted table of rooms whose status is Available
select_room()	Takes room number or type, validates availability, resolves type to a room
show_details(room)	Prints full details of the chosen room
book_room()	Confirms selection, collects guest details and nights, marks room Booked
choose_services()	Menu loop to add services until the user finishes
bill()	Calculates room charge, services, GST and total; prints the bill

6.5 Flowchart (Text Form)

```
Start
|
Show available rooms
|
Select room (number / type) --invalid--> retry
|
Show details -> Select this room? --no--> back to room list
| yes
Enter guest name, phone, guest count, nights
|
Add services? -> service menu loop
|
Special request?
```

```

|
Generate bill (room + services + 18% GST)
|
Select payment method
|
Confirm booking? --no--> room set Available -> Booking cancelled
| yes
Booking confirmed -> End

```

7. Implementation Details

7.1 Room Display

Uses f-string alignment (for example {room:<8}) with fixed widths so table columns stay aligned. Only rooms with status Available are printed.

7.2 Room Selection

- If the input matches a room number and the room is available, it is returned.
- If it matches a booked room, an error message is shown.
- Otherwise the program searches for available rooms of that type. One match is returned directly; multiple matches prompt the user to pick an exact room number.
- Input is processed with strip() and lower(), so case and extra spaces do not matter.

7.3 Guest Handling

The number of guests must be between 1 and the room's capacity, enforced with a while loop and try/except ValueError. The primary guest is recorded first, then extra guest names are collected with a for loop. Nights must be at least 1.

7.4 Services

Users may enter a number or a name. Each valid choice is appended to the services list; Finish exits the loop.

7.5 Billing

```

room_charge = price x nights
subtotal    = room_charge + service_charge
gst         = subtotal x 0.18
total       = subtotal + gst

```

7.6 Payment and Confirmation

The user picks Cash, UPI, or Card. A summary is shown, and on final yes the booking is confirmed; otherwise the room status is reset to Available.

8. Sample Run

```

Enter room number or room type: double
Available rooms of type 'Double': 103, 104
Please specify the exact room number: 103
Select this room? (yes/no): yes

```

```

Enter primary guest name: Rahul
Enter contact number: 9876543210
Enter total number of staying guests (Max 2): 2
Enter guest 2 name: Priya
Enter number of nights: 2
Add hotel services? (yes/no): yes
Enter service number or name: 1
Enter service number or name: 5

Room Charge : Rs. 3600
Services     : Rs. 250
Subtotal     : Rs. 3850
GST (18%)    : Rs. 693.0
TOTAL        : Rs. 4543.0

```

9. Concepts Applied

- Dictionaries and lists (nested list values)
- User-defined functions and return values
- while loops with break and return
- for loops and if/elif/else
- Exception handling (try/except)
- String methods (strip, lower, capitalize, join)
- Formatted output with f-strings
- Recursion (book_room() calls itself if the room is rejected)

10. Testing

Test Case	Input	Expected Result	Result
Valid room number	101	Room details shown	Pass
Room type, multiple rooms	double	Asks for exact room number	Pass
Invalid room	999	Room or available room type not found	Pass
Guests above capacity	5 (Single)	Error, asks again	Pass
Non-numeric guests/nights	abc	Please enter a valid number	Pass
Zero nights	0	Error, asks again	Pass
Invalid service/payment	xyz	Invalid choice	Pass
Cancel at confirmation	no	Room becomes Available again	Pass

11. Limitations

- Only one booking per run
- No data persistence (file or database)
- No check-in/check-out dates or calendar availability

- Phone number and names are not validated
- GST is applied uniformly at 18%
- Special request is stored but not used further

12. Future Scope

- Save and load bookings using files, CSV, or SQLite
- Date-based availability and check-out
- Multiple bookings per session, cancellation by booking ID
- Admin panel to add or edit rooms and prices
- GUI (Tkinter) or web version (Flask/Django)
- Printable invoice as PDF

13. Conclusion

The project successfully automates room booking and billing using core Python. It prevents double booking, validates input, and produces accurate itemised bills. It also builds practical skills in program structure, data handling, and input validation, and can be extended into a fuller hotel management system.

14. References

- Python Documentation: <https://docs.python.org/3/>
- Python Tutorial, Input and Output (f-strings):
<https://docs.python.org/3/tutorial/inputoutput.html>